Visions PV .035 · bench protocol for T2-E1 – T2-E4

IVUS Tier 2 / Tier 3 Acceptance Protocol — Visions PV .035

This document is the **bench protocol** for the experiments that supply the sim-vs-real reference data consumed by Tier 2 and Tier 3 of the simulator acceptance criteria (../../i4h-sensor-simulation/docs/simulation_acceptance_criteria.md). Tier 1 calibration of the probe itself is covered separately in ivus_calibration_protocol.md; this protocol does **not** repeat that work and assumes the calibration sheet at `../p035_visions/volcano_s5i.yaml` (with its provenance in `../p035_visions/parameter_sheet.csv` and outstanding bench requests in `../p035_visions/calibration_delta.md`) has already been populated for the probe under test.

Why these four experiments

The protocol is organised as **four bench experiments**, **T2-E1 — T2-E4**, mirroring §2.2 of the AC document. The experiment IDs use the `T2-` prefix to avoid collision with the Tier 1 calibration `E1–E9` numbering used in the calibration protocol cited above.

ID	Phantom	What it characterises	Annotations
T2-E1	Uniform lumen-medium tank	Lumen-medium acoustic statistics (histogram, Rayleigh σ, mean vs depth)	None
T2-E2	Wall-material tube in lumen medium	Wall-material acoustic statistics (histogram, texture), **and** soft-tissue interface reflectivity at the inner / outer wall surfaces, **and** wall attenuation (from inner-vs-outer-wall amplitude ratio)	None
T2-E3	Bulk-tissue slab (Syndaver muscular pad imaged away from any vessel, or butcher muscle)	Bulk-tissue acoustic statistics (histogram, texture)	None
T2-E4	Vessel-with-surroundings pullback (Syndaver muscular vessel pad)	Whole-frame deployment-shaped scenes for Tier 3 (FID + inference parity); Tier 2 visual review	Lumen + intima contours per frame on the Tier 3 split

Each experiment includes:

1. Purpose — which AC rows and substrate-sheet entries it produces.
2. Equipment list (with tolerance thresholds).
3. Console / device settings.
4. Step-by-step procedure.
5. Data to acquire (file naming and contents).
6. Analysis — formulas and routines used to extract each metric.
7. Acceptance criteria (the bench-side QC checks; the simulator-vs-bench comparison thresholds are in §A.2.3).

> All experiments are run with the same probe (Visions PV .035 / Volcano s5i), the same console, and at the **same operating point** (gain, imaging diameter, TGC) as the calibration-sheet reference operating point. Tank and phantom temperature is held at **22 ± 1 °C** unless the substrate parameter sheet specifies otherwise. Speed of sound in degassed water at 22 °C is taken as **c_water = 1488 m/s** (Marczak 1997).

Common equipment

Item	Spec	Tolerance / Notes
Probe + console	Visions PV .035 (10 MHz peripheral IVUS), Volcano s5i	Same unit and same catheter S/N for all experiments
Water tank	≥ 300 × 200 × 150 mm, non-reflective lining	Walls ≥ 30 mm from any catheter or phantom interface in any imaging direction
Degassed deionized water	Dissolved O_2 ≤ 4 ppm	Let sit ≥ 24 h after degassing; T = 22 ± 1 °C
Calibrated thermometer	± 0.2 °C	Log T at the start and end of every session
Catheter mount / fixture	Rigid clamp, axial alignment ≤ ±0.1 mm	Holds the probe horizontal; does not occlude the imaging plane
Frame capture (DICOM, primary)	Same export pipeline as `P_035_PointScatter/` (DICOM with private tags for imaging diameter, gain, TGC)	One DICOM file per acquisition; per-frame metadata via private tags as in the existing dataset
Frame capture (raw RF or polar `.npy`, preferred where available)	DAQ-based RF tap or vendor polar export	Optional; if available, capture in addition to DICOM, not instead
Photo / video record of mount	Smartphone-quality is acceptable	Each session must include a photograph of the phantom mounting and any deviations from the protocol

> **Data format.** The default and required output of every experiment in this protocol is **DICOM**, matching the existing `P_035_PointScatter/` dataset (`FILE0000`, `FILE0001`, …). Raw RF or vendor polar `.npy` exports are welcome **in addition to** DICOM if the lab can produce them, but they are not required and the protocol's analysis pipeline assumes DICOM is available.

Common console settings (reference operating point)

These are the **explicit device-side settings** the lab applies on the s5i console for every acquisition in this protocol. They match the operating point at which the existing `P_035_PointScatter/` calibration dataset was acquired, which is the only operating point at which the simulator's calibration sheet (`../p035_visions/volcano_s5i.yaml`) is currently valid.

Setting	Value	Notes
Imaging mode	IVUS B-mode	—
Catheter	Visions PV .035	The calibration sheet is for this catheter family only.
Imaging diameter	**60 mm**	Produces a 0.12 mm/pixel display pitch (the calibration-dataset value). Imaging diameters of 35 mm (0.07 mm/pixel) or 40 mm (0.08 mm/pixel) are allowed *only* if Tier 1 has been re-run at the chosen diameter and the calibration sheet has been re-validated against it.
Gain slider	**54**	The simulator's reference gain.
TGC sliders	**All sliders at the centre detent.**	The calibration dataset was acquired with operator TGC effectively flat (≤ ~2 dB applied at any depth in water). Photograph the slider state at the start of every session and include the photo in the metadata; if the slider hardware does not have a centre detent, set every slider to its mid-position by visual inspection and photograph.
Acoustic Reference (AR)	**ON**	The calibration dataset was acquired with AR enabled (DICOM private-tag confirmed).
Speckle reduction / smoothing	**OFF**	—
Compounding	**OFF**	—

If any setting deviates from the values above, **stop**: the experiment as captured will not be comparable to the sim-side configuration and Tier 1 must be re-validated at the deviated setting before proceeding.

Substrate identification

Every experiment that uses a lumen medium, wall material, or bulk tissue must record the substrate's **identity and batch** in the session log. The substrate parameter sheet (§A.2.1) of the AC document is the canonical store; `substrate_id` strings used in filenames must match the `Identity` field there.

Recommended `substrate_id` patterns:

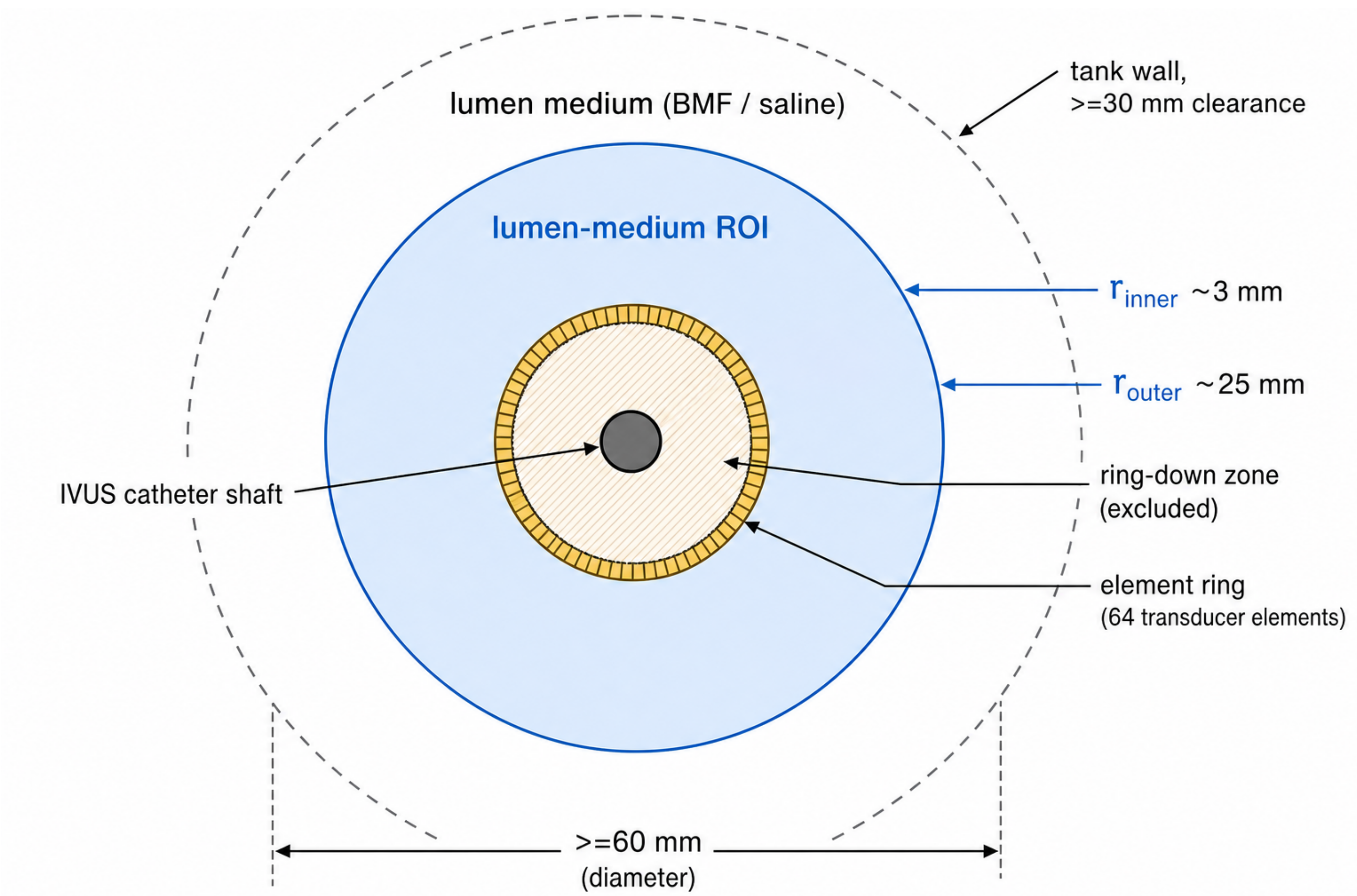
- Lumen medium: `bmf-ramnarine-batch-<YYYYMMDD>` for Ramnarine BMF; `bmf-cirs-046-lot-<###>` for CIRS commercial BMF; `saline-09pct` if saline is used (logged but note the limitation in §T2-E1 below).
- Wall material: `syndaver-customvessel-<part>-lot<###>` (the custom Syndaver iliac / upper-leg vessels); `butcher-<species>-<vessel>-batch-<YYYYMMDD>` if a butcher source is used.
- Bulk tissue: `syndaver-pad-bulk-lot<###>` (the Syndaver muscular vessel pad bulk, away from any vessel); `butcher-<species>-skeletal-batch-<YYYYMMDD>`.

Recommended phantom-material recipes / sources

The substrate parameter sheet entries for backscatter and attenuation are ideally measured on the actual batch (substitution method — Appendix A). If a published recipe or commercial source is used, the following are the recommended starting points for diagnostic-frequency (≤ 10 MHz) ultrasound. Detailed preparation steps are in **Appendix B**.

- **Blood-mimicking fluid (BMF):** Holdsworth Lab BMF SOP (Robarts Research Institute), adapted from Ramnarine et al. 1998. CIRS Model 046 is the turnkey commercial alternative. See Appendix B.1.
- **Tissue-mimicking material (TMM, optional — only required if the Syndaver muscular vessel pad bulk or butcher skeletal muscle is unavailable for T2-E3):** IEC 61685 / Teirlinck et al. 1998 agar TMM. See Appendix B.2.

T2-E1 — Uniform lumen-medium tank



Produces: Tier 2 lumen rows (palette histogram, Rayleigh σ in lumen medium, lumen-medium mean palette vs depth) per §A.2.3, and the bench-side characterization data that populates the **lumen-medium** group of the substrate parameter sheet (§A.2.1).

> **Goal.** Isolate the probe's representation of the lumen medium. No vessel, no wall material, no scatterer at depth other than the medium itself. The acquisition is a long, uniform A-line through the chosen fluid.

Equipment

Item	Spec	Tolerance
Lumen medium under test	Ramnarine BMF (recipe above) or CIRS Model 046 BMF	Recipe / commercial source documented; batch ID logged; temperature held at 22 ± 1 °C; allow ≥ 30 min equilibration after pouring before capture
Tank	Common tank above; entirely filled with the lumen medium under test	All probe-to-wall distances ≥ 30 mm in the imaging plane and ≥ 30 mm along the catheter shaft from the element ring
Catheter mount	Common; probe horizontal, element ring at the centre of the tank	≤ ±0.1 mm axial position

Console settings

Reference operating point (table above). No deviation.

Procedure

1. Fill the tank with the lumen medium under test. If the medium is freshly mixed (e.g. Ramnarine BMF), allow ≥ 30 min equilibration; visually confirm no settling layer.
2. Measure and log the temperature.
3. Mount the catheter horizontally; insert the tip ≥ 50 mm into the tank with the element ring near the centre of the tank, ≥ 30 mm clearance to all walls.
4. Set the console to the reference operating point. Verify the gain slider, imaging diameter, TGC, and acoustic-reference state match the calibration sheet (record any digital readout or screen capture).
5. Capture ≥ **60 frames** in static position at the reference operating point. (60 gives margin for any per-frame artifacts.)
6. Photograph the tank, catheter mount, and console settings screen.

Data to acquire

- `t2_e1_<substrate_id>.dcm` — DICOM volume containing the captured frames (one DICOM per acquisition is fine; multiple if the console splits captures).
- `t2_e1_<substrate_id>_metadata.json` — gain slider, imaging diameter, TGC schedule, acoustic-reference state, frame rate, T_water, substrate identity and batch, fluid temperature at start and end, notes on equilibration / agitation, paths to photos.
- *(Optional, preferred where available)* `t2_e1_<substrate_id>_polar.npy` — vendor polar-domain export, shape `(N_frames, N_az, N_r)`.

Analysis

Implementation lives in `instrument-calibration/p035_visions/` (one extract script per experiment is recommended; place under `extract_t2_e1_lumen_medium.py`).

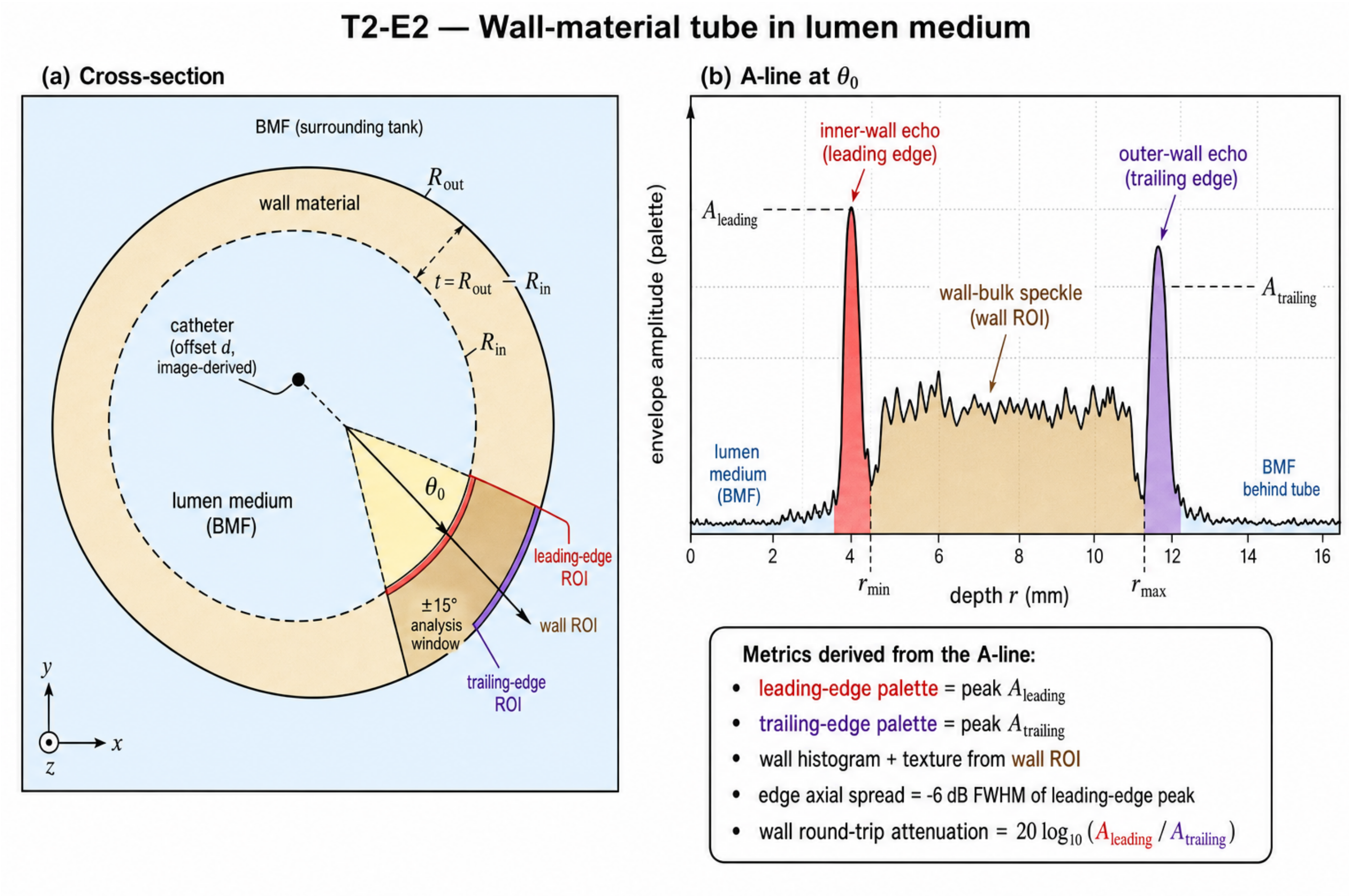
Output	Method
Lumen-medium ROI mask	Per §B.1: polar pixels with r in `[r_inner, r_outer]`, where `r_inner` is just outside the calibration sheet's ring-down extent and `r_outer` is the deepest sample free of tank-wall returns; band must span ≥ 5 mm.
Palette histogram	Per-ROI pixel histogram, 1 palette unit per bin.
Rayleigh σ	Invert log compression to RF amplitude using `volcano_s5i.yaml`'s `log_floor` and `log_multiplier`; fit `scipy.stats.rayleigh.fit` per §B.3.
Mean palette vs depth	Bin pixels into 1 mm radial bins inside the ROI, take the mean per bin; report as a 1-D vector aligned to depth.
Lumen-medium attenuation (substrate sheet input, optional)	Substitution method (Appendix A) on the **same tank** during the same session if a reference reflector is set up; else cite the recipe / commercial source's published value at 10 MHz with explicit uncertainty.

Bench acceptance criteria

These are the bench-side QC checks; sim-vs-bench thresholds are in §A.2.3.

- ROI band length ≥ 5 mm and free of tank-wall echoes (visual check).
- Per-frame mean palette in the ROI stable to within ±2 palette units across the 60-frame capture (no thermal or settling drift).
- Substrate identity, batch, and temperature recorded; if any mid-session parameter changes, segment the capture and treat halves as independent.

T2-E2 — Wall-material tube in lumen medium



Produces: the Tier 2 wall rows (palette histogram, lateral autocorrelation length, GLCM statistics), the Tier 2 interface rows (leading-edge palette at the lumen↔wall surface, trailing-edge palette at the wall↔lumen surface, edge axial spread) and the wall-attenuation input to the substrate parameter sheet — all from the **same** tube acquisition, by analysing different parts of the polar B-mode within the near-normal-incidence azimuthal sector.

> **Goal.** Isolate the probe's representation of the wall material at the depth band the wall occupies in deployment, and quantify the reflection at the lumen↔wall and wall↔lumen interfaces. A tube is used (not a slab) because all available wall material is in tube form (custom Syndaver iliac / upper-leg vessels, optionally butcher arteries). The lab does **not** need to physically measure the catheter offset inside the tube; all geometry is recovered from the acquired polar images by segmentation post-acquisition.

Equipment

Item	Spec	Tolerance
Wall material	One of (priority order): (a) custom Syndaver vessel material (iliac / upper-leg set), (b) butcher arterial wall	Material identity and batch logged; uniform wall thickness over the imaging length; no calcified plaque, no perforation
Tube	Inner radius `R_in` and outer radius `R_out` measured by digital callipers **before mounting**	`R_in` and `R_out` ± 0.05 mm; wall thickness `t = R_out − R_in` ± 0.1 mm uniform along length
Surrounding fluid	The same lumen medium used in T2-E1 (Ramnarine BMF / CIRS BMF) on **both** sides of the wall	Fluid identity, batch, and temperature logged
Phantom mount	Holds the wall tube rigid in the imaging plane, parallel to the catheter long axis. The catheter passes through the tube.	The catheter must not contact the wall, but its **exact** offset from the tube centre does not need to be controlled or physically measured — set it up so the catheter is offset enough from the centre that one side of the tube is closer than the other (at least 0.5 mm inside the tube, but not touching the wall).

Console settings

Reference operating point.

Procedure

1. Mount the wall tube in the imaging plane, with its long axis parallel to the catheter long axis. The tube must extend ≥ 10 mm past the element ring on both sides so the imaged wall band is fully within the tube.
2. Insert the catheter through the tube. Position it so it is offset from the tube centre but **not** in contact with the wall. The exact offset is recovered post-acquisition from the image; do not attempt to measure it physically.
3. Fill the tube and surrounding tank with the chosen lumen medium; eliminate trapped air on both sides of the wall.
4. Capture ≥ **60 frames** in static position. The catheter must remain stationary; small flow in the surrounding fluid is acceptable provided the wall does not move.
5. Photograph the mount.

Data to acquire

- `t2_e2_<substrate_id>.dcm` — DICOM, ≥ 60 frames at the reference operating point.
- `t2_e2_<substrate_id>_tube_geometry.json` — measured `R_in`, `R_out`, and wall thickness `t` from the digital-calliper measurement of the **tube** (no catheter-offset measurement required).
- `t2_e2_<substrate_id>_metadata.json` — identical fields to T2-E1 metadata, plus the wall-material identity, batch, source.
- *(Optional)* `t2_e2_<substrate_id>_polar.npy` — vendor polar-domain export.

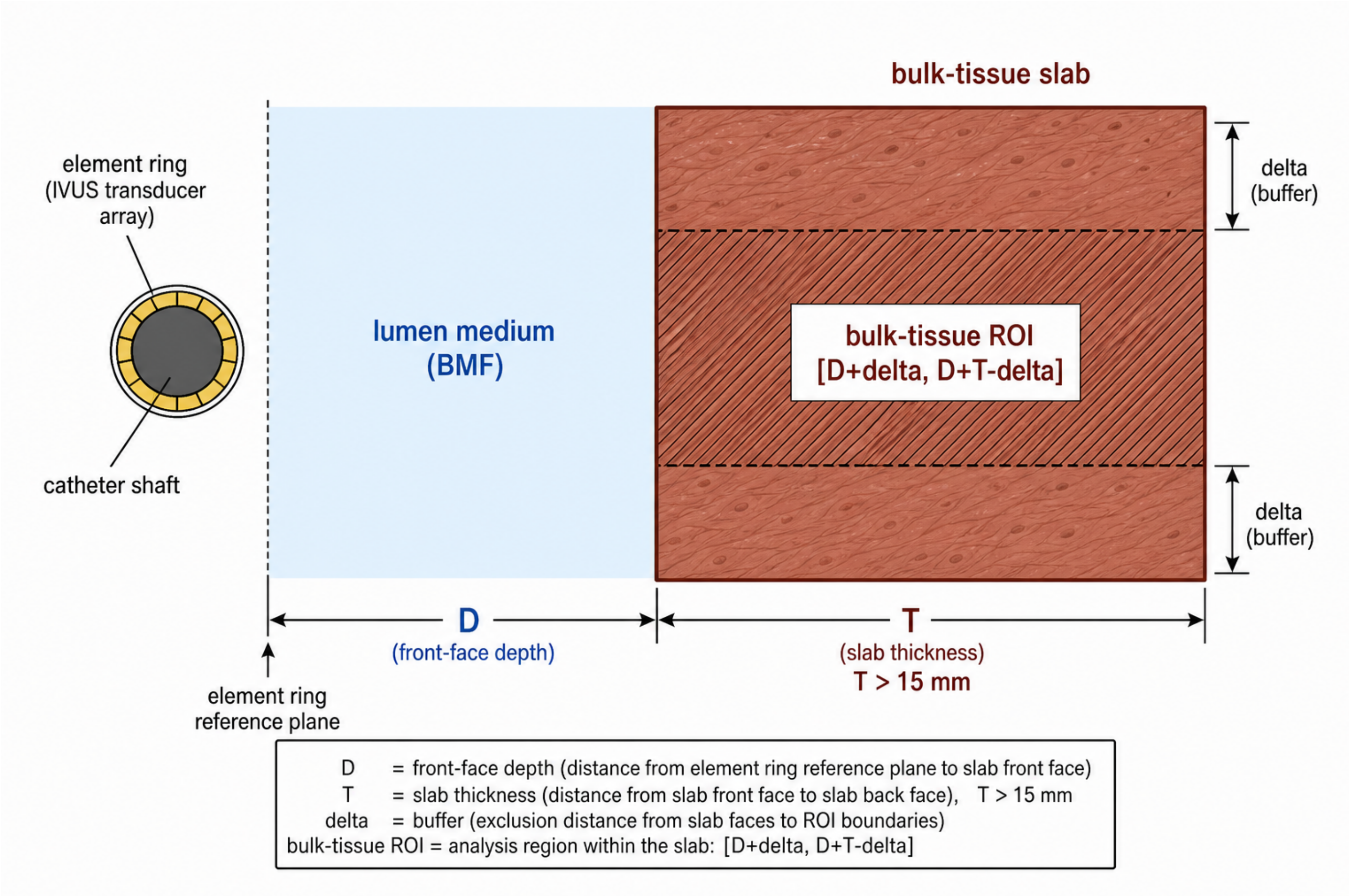
Analysis

Output	Method
Catheter offset `d` and near-normal-incidence azimuth θ_0	Image-based segmentation: in each polar frame, fit the inner-wall echo as a function of azimuth — the azimuth at which the inner-wall depth is minimum (`r_min`) gives θ_0; offset `d = (R_in − r_min)`. Report mean and σ across frames.
Wall ROI mask	Per §B.1: annular polar region between the segmented inner-wall and outer-wall echoes, restricted to angles within ±15° of θ_0 (near-normal incidence).
Wall palette histogram	Per-ROI pixel histogram, 1 palette unit per bin.
Lateral autocorrelation length	Per §B.4.1; mean across radial bins inside the wall ROI.
GLCM statistics	Per §B.4.2; per-patch contrast, homogeneity, energy, correlation across the session.
Leading-edge ROI	Narrow band `[r_min − w, r_min + w]` at the inner-wall echo, restricted to angles within ±15° of θ_0, with `w ≥ 1.5 ×` the Tier 1 axial PSF FWHM at `r_min`.
Leading-edge palette	Mean over angles and frames of the peak palette in the leading-edge ROI.
Trailing-edge ROI	Narrow band `[r_max − w, r_max + w]` at the outer-wall echo, where `r_max = r_min + t` is the segmented outer-wall depth at θ_0.
Trailing-edge palette	Mean over angles and frames of the peak palette in the trailing-edge ROI.
Edge axial spread	Axial –6 dB FWHM of the leading-edge bright band on the A-line trace, averaged over the analysis-window angles.
Predicted leading-edge palette	Predicted reflection coefficient `R = (Z_wall − Z_fluid) / (Z_wall + Z_fluid)` from substrate-sheet impedances; convert reflected envelope amplitude to palette using `volcano_s5i.yaml`'s `log_floor` and `log_multiplier`.
Wall round-trip attenuation (dB)	`α_round-trip = 20 · log10(A_leading / A_trailing)`, where `A_leading` and `A_trailing` are the un-log-compressed envelope amplitudes from the leading- and trailing-edge ROIs. The leading and trailing impedance contrasts are equal (fluid–wall on both faces), so the amplitude ratio is dominated by the round-trip wall attenuation `2 · α_wall · t`. Solve for the wall's one-way attenuation `α_wall` in dB/cm.
Wall acoustic impedance and attenuation (substrate sheet inputs)	`α_wall` from the row above. Impedance from leading-edge palette inverted through the predicted-reflection formula.

Bench acceptance criteria

- Tube `R_in` and `R_out` measurements reproducible across two callipering sessions to within ±0.05 mm.
- Catheter does not contact the wall in any frame (segmented inner-wall depth `r_min ≥ 0.3 mm` everywhere along the capture).
- No trapped air in the tube interior or tank (visual check + B-mode check for high-amplitude near-field artefacts that would indicate bubbles).
- Inner-wall echo SNR ≥ 20 dB above the surrounding-fluid speckle floor in the analysis window.
- Outer-wall echo SNR ≥ 10 dB above the surrounding-fluid speckle floor (lower threshold because the wall attenuates the round trip).

T2-E3 — Bulk-tissue slab



Produces: Tier 2 bulk-tissue rows (palette histogram, lateral autocorrelation length, GLCM statistics) per §A.2.3, and the bench-side characterization data that populates the **bulk tissue** group of the substrate parameter sheet.

> **Goal.** Isolate the probe's representation of bulk tissue at the radial depths a perivascular ROI would occupy in deployment. The Syndaver muscular vessel pad imaged **away from any vessel** is the primary phantom — the bulk material is the same as the perivascular material in T2-E4, so the same pad and the same session covers both.

Equipment

Item	Spec	Tolerance
Bulk-tissue slab	One of (priority order): (a) Syndaver muscular pad bulk (away from any vessel — same pad as T2-E4), (b) butcher skeletal muscle, (c) Madsen / IEC 61685 TMM	Slab thickness ≥ 15 mm so the back face is beyond `t_far`; no fascia, gristle, or fat seams in the imaging volume; surface flatness within 1 mm peak-to-valley
Surrounding fluid (in front of slab)	Same lumen medium as T2-E1 / T2-E2	Identity, batch, temperature logged
Phantom mount	Slab rigidly held with its surface parallel to the catheter long axis	Approximately parallel; exact orientation recovered from the segmented front-face echo post-acquisition

Console settings

Reference operating point.

Procedure

1. Mount the bulk-tissue slab so its front face is approximately parallel to the catheter long axis at a comfortable working distance from the element ring (5–10 mm front-face depth is typical). The exact depth is recovered from the acquired image post-acquisition and does not need to be physically pre-measured.
2. Fill the tank with the chosen lumen medium up to and over the slab (slab fully immersed, no air in the imaging path).
3. Capture ≥ **60 frames** in static position.
4. Photograph the mount.

Data to acquire

- `t2_e3_<substrate_id>.dcm` — DICOM, ≥ 60 frames at the reference operating point.
- `t2_e3_<substrate_id>_metadata.json` — gain, TGC, acoustic-reference state, T_water, slab identity / batch / source / handling notes.
- *(Optional)* `t2_e3_<substrate_id>_polar.npy` — vendor polar-domain export.

Analysis

Output	Method
Slab front-face depth `D` and orientation	Image-based segmentation of the front-face echo in each polar frame; report `D` as the mean depth across the analysis-window angles, and the slab orientation relative to the catheter as the slope of the front-face echo across angles.
Bulk-tissue ROI mask	Per §B.1: polar pixels in `[D + δ, D + T − δ]` with δ ≥ 0.5 mm. `T` is the slab thickness (≥ 15 mm).
Bulk-tissue palette histogram	Per-ROI pixel histogram.
Lateral autocorrelation length	Per §B.4.1; mean across radial bins inside the ROI.
GLCM statistics	Per §B.4.2; per-statistic distributions across the session.
Bulk-tissue acoustic impedance	Estimate from the leading-edge specular reflection coefficient (front face of the slab) measured in this experiment and the lumen-medium impedance from T2-E1 / substrate sheet.
Bulk-tissue attenuation (substrate sheet input, optional)	Substitution method on a thin replicate of the slab if available, else cite literature for the material at 10 MHz.

Bench acceptance criteria

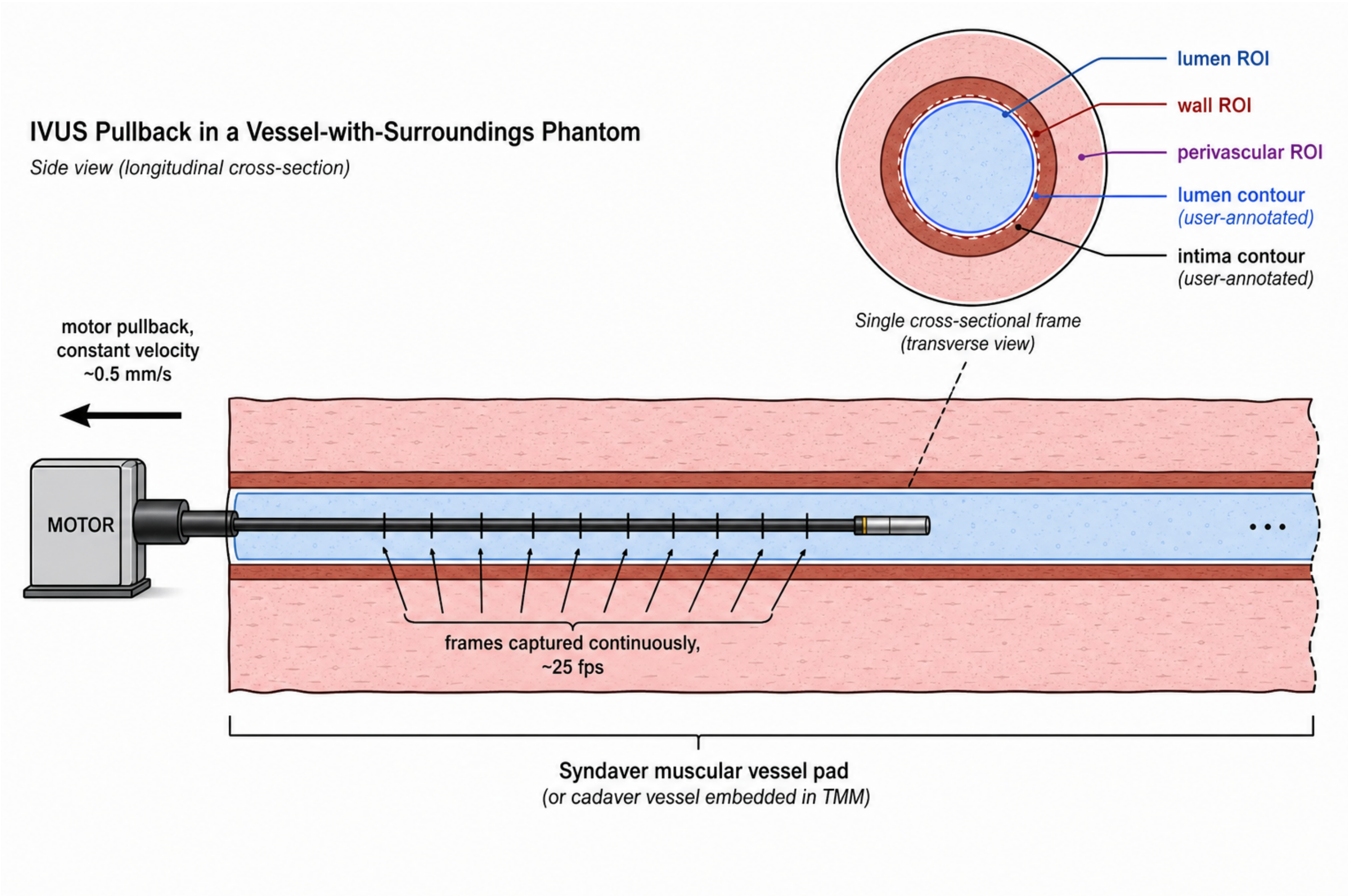
- Slab front-face echo visible across all imaging angles (slab covers the full FOV in the imaging plane).
- No trapped air at the front face of the slab.
- Slab thickness ≥ 15 mm such that the back face is at or beyond `t_far`.

Through-wall consistency check (uses T2-E2 + T2-E3 + T2-E4 jointly, no extra capture)

Once T2-E2 (wall attenuation) and T2-E3 (bulk-tissue reference) and T2-E4 (vessel-with-surroundings pullback) are all acquired, the through-wall attenuation and post-interface bulk-tissue speckle checks are computed analytically:

1. From T2-E2: extract `α_wall` (one-way attenuation, dB/cm) and the leading- and trailing-edge palette values.
2. From T2-E4: segment the perivascular ROI per §B.1; measure the wall thickness `t_E4` in the same frames.
3. Predict the perivascular palette: take the T2-E3 free-fluid bulk-tissue palette at the matching radial depth, subtract `2 · α_wall · t_E4` in dB (round-trip wall attenuation), and convert back to palette via the calibration log mapping.
4. Pass criterion: predicted vs measured perivascular palette within ±2 dB; perivascular speckle statistics (autocorrelation, GLCM) within ±30 % of T2-E3 after attenuation correction. These thresholds live in §A.2.3.

T2-E4 — Vessel-with-surroundings pullback (Phase A + Phase B)



Produces: Tier 3 reference (FID, inference parity) per §A.3, the Tier 2 visual-review row per §A.2.3, the data needed for the through-wall consistency check above, and the **primary** Tier 3 cross-distribution anchor (Phase B).

> **Goal.** Clinically-shaped IVUS pullbacks in a phantom whose lumen, wall, and perivascular tissue are all configured to match the simulator's training distribution. T2-E4 is the **only** acquisition that satisfies Tier 3.

T2-E4 is run in **two phases**, in order, on the **same** mounted vessel phantom:

- **Phase A — Visions PV .035 pullback (mandatory).** The Visions PV .035 catheter on the Volcano s5i, at the reference operating point defined above. Provides the Tier 2 visual-review reference, the Tier 3 reference, and the through-wall-consistency-check input.
- **Phase B — Cross-probe replicate (optional but strongly recommended).** A *different* IVUS probe — Boston Scientific IVUS or a smaller Volcano IVUS catheter — pulled through the **same** vessel section of the **same** Syndaver pad, at *that* probe's own reference operating point. Provides the **primary** Tier 3 cross-distribution anchor in §A.3.3 of the AC document. Without Phase B, only the Roboflow-based secondary anchor (§2.3 of the AC document) is available, and that anchor confounds "different probe" with "different anatomy".

Common phantom setup (used by both phases)

Item	Spec	Tolerance
Vessel phantom	Syndaver muscular vessel pad (peripheral-relevant 6 mm vessel preferred over 3 mm where available)	Vessel patent end-to-end; lumen length sufficient for ≥ 100 mm of pullback travel
Lumen medium	Same BMF as T2-E1 (Holdsworth / Ramnarine or CIRS)	Identity and batch logged; same temperature as the rest of the session
Lumen flow *(optional, nice-to-have)*	If a continuous-flow pump is available, run continuous (not pulsatile) flow at clinically realistic rate	Rate logged in metadata. Pulsatile flow is **not** required and is **not assumed** unless explicitly documented in the metadata.
Phantom mounting	Stable, horizontal orientation; the mount must remain undisturbed between Phase A and Phase B	Mark the phantom orientation with respect to the mount so Phase B can verify the same vessel section is imaged

The vessel phantom is mounted **once**. Phase A is run, the Visions PV .035 catheter is withdrawn without disturbing the phantom, and Phase B is run by inserting the second IVUS probe through the same lumen.

Phase A — Visions PV .035 pullback (mandatory)

Phase A — Equipment (deltas from common setup)

Item	Spec	Tolerance
Probe	Visions PV .035 (10 MHz peripheral IVUS) on the Volcano s5i	Same catheter S/N as T2-E1 / T2-E2 / T2-E3
Catheter motion	**Motorised** constant-velocity pullback (e.g. 0.5 or 1.0 mm/s) is preferred. **Manual** pullback is acceptable when a motor is unavailable.	If motorised, velocity tolerance ≤ ±5 % over the pullback length. If manual, the motion type must be explicitly recorded in the metadata; no specific velocity profile is required.
Frame capture	DICOM, continuous capture for the duration of the pullback, ≥ 25 fps	One DICOM containing the full pullback is preferred; per-frame timestamp recorded
Annotation pipeline	Lumen and intima contours per frame on the Tier 3 split (≥ 1000 frames)	Annotation tool documented; contour format compatible with the simulator's polar coordinate convention

Phase A — Console settings

The pullback is acquired with the **same operating point** as T2-E1 / T2-E2 / T2-E3 (the reference operating point in the *Common console settings* section above). Deviation invalidates the Tier 1 calibration round-trip and downstream comparisons.

Phase A — Procedure

1. Mount the vessel phantom in a stable, horizontal orientation per the *Common phantom setup* table above. Mark the phantom orientation on the mount.
2. Insert the Visions PV .035 catheter through the vessel lumen so the element ring exits the distal end of the phantom and is then withdrawn through the imaging volume during pullback.
3. Establish lumen medium fill: prime the lumen with the BMF; verify no bubbles in the imaging volume by a static B-mode check at the start of the pullback path. If a continuous-flow pump is available, document the flow rate; pulsatile flow is not required.
4. Confirm console settings match the Visions PV .035 reference operating point. Take a screen capture for the metadata.
5. Run the pullback over ≥ 100 mm of vessel length. - **Motorised (preferred):** constant velocity (0.5–1.0 mm/s typical); record the commanded velocity. - **Manual (acceptable):** smoothly withdraw the catheter at an approximately constant rate; avoid stalls. Record the motion type as `manual` in the metadata. Capture **continuously**; the total number of frames must be ≥ **2200** before any quality filtering, so that the 50 / 50 Tier 2 / Tier 3 split (§2.2.1) yields ≥ 1000 frames per half after dropping any corrupt frames.
6. Repeat pullbacks if the first attempt has identifiable transient issues (sync drops, console reset, motor stutter); discard runs that cannot be reconciled to a single continuous pullback path.
7. Run the annotation pipeline on the Tier 3 split: produce per-frame lumen and intima polar contours.

8. Photograph the entire mount.
9. Withdraw the Visions PV .035 catheter **without disturbing the phantom mount**.

Phase A — Data to acquire

- `t2_e4_<substrate_id>.dcm` — DICOM containing the full Phase A pullback.
- `t2_e4_<substrate_id>_pullback.json` — motion type (`motorised` / `manual`), motorised velocity if applicable, pullback length, start / end timestamps, frame timestamps.
- `t2_e4_<substrate_id>_metadata.json` — gain, TGC, acoustic-reference state, T_water, vessel-phantom identity (pad ID and vessel section), lumen medium identity / batch, flow setting if any, photos.
- `t2_e4_<substrate_id>_split.json` — frame-level Tier 2 / Tier 3 split index list, generated **before** any sim-vs-real comparison is run.
- `t2_e4_<substrate_id>_annotations.zip` — per-frame lumen and intima polar contours for the Tier 3 split.

Phase B — Cross-probe replicate (optional but strongly recommended)

> **Why Phase B matters.** §A.3.3 of the AC document gates the simulator on "does it look more like the Visions PV .035 than like a generic IVUS probe imaging the same anatomy?". Phase B is what makes that question cleanly answerable — the same vessel section of the same Syndaver pad with the same lumen medium, imaged by a *different* IVUS probe, isolates probe-only differences from anatomy / phantom differences. Without Phase B the gate falls back to a Roboflow-based anchor that confounds the two.

Phase B — Equipment (deltas from common setup)

Item	Spec	Tolerance
Second IVUS probe	One of (in priority order): (a) **Boston Scientific IVUS** (e.g. Opticross HD, AVANTI), (b) a **Volcano IVUS** catheter of a different size from the Visions PV .035 (e.g. Eagle Eye Platinum 20 MHz). See "Probe selection" below.	Catheter S/N, model, and manufacturer logged in Phase B metadata
Console for the second probe	The manufacturer's own console for that probe (Boston Scientific clinical console for the BSc probe, or the Volcano s5i for a different-size Volcano catheter)	Console identity and firmware version logged
Vessel phantom	**Same** Syndaver pad and **same** vessel section as Phase A; phantom mount left undisturbed between phases	Phase B images the same anatomy as Phase A; metadata explicitly records this
Lumen medium	**Same** BMF batch as Phase A	Identity and batch repeated in Phase B metadata for traceability
Catheter motion	Same options as Phase A (motorised preferred, manual acceptable)	Motion type recorded in metadata
Frame capture	DICOM, ≥ 1000 frames after quality filtering	Annotations not required

Probe selection. Phase B's value comes from imaging the *same* anatomy with an acoustically *different* probe. The Boston Scientific IVUS family and smaller / higher-frequency Volcano catheters all qualify because they differ from the Visions PV .035 in centre frequency, beam geometry, and post-processing. If multiple second probes are available, prefer the one whose imaging frequency differs most from 10 MHz (e.g. a 20 MHz Eagle Eye Platinum or a Boston Scientific 40 / 60 MHz catheter), since that maximises the "different-probe" distance the triangle test is measuring against.

Phase B — Console settings

Use **the second probe's own reference operating point**: the gain, imaging diameter, TGC, and acoustic-reference state appropriate for that probe and console. Do **not** force the Visions PV .035 settings onto the second probe — the calibration sheet at `../p035_visions/volcano_s5i.yaml` applies to the Visions PV .035 only. Whatever the second probe's operating point is, record every adjustable setting in the Phase B metadata (slider values, screen capture).

Phase B — Procedure

1. With Phase A complete and the Visions PV .035 catheter withdrawn, confirm the phantom mount is undisturbed (visual check against the orientation marks made at Phase A step 1).
2. Insert the second IVUS catheter through the **same vessel section** used in Phase A (best-effort — exact axial alignment is not required, but the catheter must enter the same vessel and traverse the same approximate length).
3. Confirm the lumen is still primed with BMF; top up if necessary. Verify no bubbles by a static B-mode check on the second probe.
4. Set the second probe's console to its own reference operating point. Photograph the slider state and the console-settings screen.
5. Run a pullback over the same vessel length covered in Phase A. Capture **continuously**; ≥ 1000 frames after quality filtering is sufficient. Motorised or manual pullback both acceptable; document the motion type.
6. Photograph the second-probe console and cabling.

Phase B — Data to acquire

- `t2_e4_<substrate_id>_xprobe_<probe_id>.dcm` — DICOM containing the full Phase B pullback. `<probe_id>` is a short string identifying the probe family / model, e.g. `bsc-opticross-hd`, `volcano-eep20`, etc.
- `t2_e4_<substrate_id>_xprobe_<probe_id>_metadata.json` — second probe identity / model / S/N, console identity / firmware, gain / TGC / imaging diameter / acoustic-reference state at the second probe's reference operating point, T_water, motion type and velocity if applicable, photos. Repeats the lumen-medium identity and the Phase A `t2_e4_<substrate_id>` reference for traceability.

No annotations are required for Phase B; Tier 3 only uses Phase B frames for FID computation.

Analysis

For Tier 3, see §A.3.2. The bench-side analysis for T2-E4 is limited to:

Output	Method
Frame quality filter (Phase A)	Drop frames with sync errors, console reset markers, or visibly corrupt DICOM. The drop list is logged in the split JSON.
Tier 2 / Tier 3 split (Phase A)	First contiguous 50 % of clean frames → Tier 2 visual-review reference; remaining 50 % → Tier 3 reference. Splits are generated before any sim-vs-real test is run and are not modifiable thereafter.
Visual review (informal, A.2.3)	20 frames (10 sim, 10 real) drawn from the Tier 2 visual-review split of Phase A; reviewers are blind to source.
Frame quality filter (Phase B, if collected)	Same as Phase A; logged separately. Phase B is not split — the entire clean frame set is used as the Tier 3 cross-probe anchor.

Cross-probe triangle anchor in Tier 3

§A.3.3 requires the simulator to look more like the Visions PV .035 probe than like a different probe. With Phase B collected, the triangle test has a clean anchor:

- **Primary:** `FID(sim, T2-E4 Phase A Visions PV .035 reference) ≤ FID(sim, T2-E4 Phase B cross-probe replicate)` — same phantom, same anatomy, only the probe differs.
- **Secondary:** `FID(sim, T2-E4 Phase A Visions PV .035 reference) ≤ FID(sim, Roboflow IVUS-2)` — kept as a sanity check on more diverse anatomy.

If Phase B is not collected, only the secondary (Roboflow-based) anchor is available; that anchor confounds probe with anatomy and the gate's diagnostic value is reduced accordingly.

Bench acceptance criteria

Phase A (mandatory):

- ≥ 2200 frames captured before quality filtering.
- ≥ 1000 frames per Tier 2 / Tier 3 split after quality filtering.
- ≥ 95 % of split frames carry valid lumen and intima annotations.
- Pullback motion type explicitly recorded; if motorised, velocity stable to within ±5 % across the run; no frame-time jumps > 2 frame periods.
- No air bubbles or annotation gaps lasting > 3 consecutive frames.

Phase B (if collected):

- ≥ 1000 frames captured after quality filtering.
- Phase B images the **same** vessel section of the **same** Syndaver pad as Phase A; metadata explicitly references the Phase A `<substrate_id>`.
- Phantom mount left undisturbed between Phase A and Phase B (verified by the orientation marks placed at Phase A step 1).
- Phase B's metadata explicitly records the second-probe model, S/N, console identity / firmware, and the full slider state of that probe's reference operating point. No DICOM cross-mixing between Phase A and Phase B output files.

Appendix A — Substitution attenuation measurement (optional)

For substrate-parameter-sheet entries that require a measured attenuation in dB/cm at 10 MHz (lumen medium, bulk tissue), the substitution method below is the recommended bench protocol. Wall attenuation is recovered directly from T2-E2 (inner-vs-outer-wall amplitude ratio) and does not need a separate substitution capture.

1. Mount a flat, polished metal reflector (the Tier 1 E1 reflector is suitable) at a known radial distance `D_ref` from the element ring in the chosen lumen medium with the test substrate **absent** from the imaging path.
2. Capture ≥ 30 frames; record the peak A-line palette at the reflector depth and convert to envelope amplitude `A_without` via the calibration log mapping.
3. Insert the test substrate of known thickness `L` into the imaging path in front of the reflector. Re-capture ≥ 30 frames; record `A_with`.
4. Compute attenuation: `α (dB/cm) = 20 · log10(A_without / A_with) / (2 · L · 0.1)`.

Document `L`, both peak palettes, T_water, and substrate identity / batch in the same metadata file as the parent experiment.

Appendix B — Material preparation references

When the lab prepares its own BMF or TMM, the published SOPs / recipes below are the canonical starting points. This protocol does **not** reproduce the full step-by-step preparation; the lab follows the cited SOP and records the batch and preparation details in the experiment metadata.

B.1 Blood-mimicking fluid (BMF)

Recommended SOP. Holdsworth Lab BMF SOP (Robarts Research Institute), `making_bmf_sopt.pdf`. The Holdsworth SOP is adapted from Ramnarine, Nassiri, Hoskins & Lubbers, *"Validation of a new blood-mimicking fluid for use in Doppler flow test objects"*, Ultrasound in Medicine and Biology 24(3):451–9 (1998), and characterises the resulting fluid for diagnostic ultrasound frequencies. Suitable as the lumen medium for T2-E1, T2-E2, and T2-E4 in this protocol.

Commercial alternative. CIRS Model 046 BMF — use as supplied; record lot number in the experiment metadata. No in-house preparation needed.

What to log per batch (in addition to the SOP's own log fields).

- Batch ID matching the `substrate_id` pattern in this protocol's *Substrate identification* section (e.g. `bmf-ramnarine-batch-20260507`).
- Date prepared, operator, ingredient lot numbers and weights.
- Storage conditions (container, temperature, since when).
- Any in-house acoustic measurements made on the batch (substitution attenuation per Appendix A; speed-of-sound spot check if a calibrated through-transmission rig is available).
- Photographs of the batch as poured / in storage.

B.2 Tissue-mimicking material (TMM)

Recommended SOP. Souza, Santos, Oliveira, Alvarenga & Costa-Felix, *"Standard operating procedure to prepare agar phantoms"*, J. Phys.: Conf. Ser. 733 012044 (2016), open-access PDF at `iopscience.iop.org/article/10.1088/1742-6596/733/1/012044/pdf`. The Inmetro (Brazilian National Institute of Metrology) SOP gives a step-by-step preparation procedure for the IEC 60601-2-37 agar TMM (glycerol 11.21 %, deionised water 82.95 %, benzalkonium chloride 0.47 %, SiC –400 mesh 0.53 %, Al_2O_3 0.3 µm 0.88 %, Al_2O_3 3 µm 0.94 %, agar 3.02 % by weight) with explicit handling of the points the IEC standard leaves under-specified (stirring method, water-bath temperature window, cooling temperature before pouring, storage solution). This is the recommended SOP for in-house TMM preparation.

Underlying standards and characterisation references.

- IEC 60601-2-37 / IEC 61685 — the source recipe.
- Teirlinck, Bezemer, Kollmann *et al.*, *Ultrasonics* 36 (1998) 653–660 — original characterisation of the IEC TMM at diagnostic frequencies.
- Inglis *et al.*, *UMB* 44(2):440–51 (2018) — per-component attenuation breakdown for varied recipes.
- Cannon *et al.*, *UMB* 41(2):574–82 (2014/2015) — characterisation up to 60 MHz (relevant if the lab pushes the TMM beyond 10 MHz).

When to prepare TMM. Only prepare TMM in-house if the Syndaver muscular vessel pad bulk and butcher skeletal muscle are unavailable for T2-E3, **or** if the lab needs a TMM characterised for the substrate parameter sheet at a level of certainty above what literature provides.

What to log per batch. Same as B.1 above, plus the SOP step at which any deviation occurred (water-bath temperature, stirring rpm profile, cooling-before-pour temperature) — the Souza SOP is explicit that these points drive batch-to-batch variability.

B.3 Substrate batch logging convention

For every BMF or TMM batch — whether prepared in-house or commercial — the batch's `substrate_id` must match the `Identity` field of the substrate parameter sheet (§A.2.1) of the AC document. If a batch is used across multiple experiments (e.g. one BMF batch covers T2-E1, T2-E2, and T2-E4), the same `substrate_id` is reused in every experiment's metadata so the AC analysis pipeline can group acquisitions by batch.